

Lemma. Let G be a group and A be a set.

For an Action of G on A , $|G \cdot G_a| = |G_a|$

Proof. Since for any $g_1, g_2 \in G_a$, $g_1 g_2^{-1} a = g_1 a = a$, so $g_1 g_2^{-1} \in G_a$.

Thus G_a is a subgroup. And, define a map

$$G_a \rightarrow \{gG_a \mid g \in G\}; g \cdot a \mapsto gG_a$$

Then, $g_1 G_a = g_2 G_a$ implies $g_1^{-1} g_2 \in G_a$, or $g_1^{-1} g_2 a = a$, or $g_1 a = g_2 a$.

So injective, and the surjective is trivial. Thus, Proved.

Thm. Let G be a group and H be a subgroup of G .

An action $G \times \{gH \mid g \in G\} \rightarrow \{gH \mid g \in G\}; (g_1, g_2 H) \mapsto g_1 g_2 H$ satisfies:

1) It acts transitively. (It is clear, since $\{gH \mid g \in G\}$ is a partition of G).

2) $G_H = H$

3) $\ker \pi_H = \bigcap_{x \in G} xHx^{-1}$ is the largest normal subgroup of G contained in H .

Proof. $\ker \pi_H = \{g \in G \mid g x H = x H \text{ for all } x \in G\}$
 $= \{g \in G \mid x^{-1} g x H = H \text{ for all } x \in G\} = \bigcap_{x \in G} x H x^{-1}$

Let $g \in G$ and $h \in \ker \pi_H$. Then, for any $x \in G$,

$$x^{-1}(g h g^{-1})x = (g^{-1} x)^{-1} \cdot h \cdot (g^{-1} x) \in \ker \pi_H, \text{ so } \ker \pi_H \trianglelefteq G.$$

And, given $g \in \ker \pi_H$ and $h \in H$, $h^{-1} g h H = H \Rightarrow g \in H$, so $\ker \pi_H \leq H$.

Finally, if $N \leq H$ and $N \trianglelefteq G$, then given $x \in G$

$$N = x N x^{-1} \leq x H x^{-1}, \text{ so } N \subseteq \bigcap_{x \in G} x H x^{-1}$$

Corollary. If G is a finite group of order n and p is the smallest prime dividing n , then any subgroup of index p is normal.

Proof. Let $H \leq G$ such that $|G:H| = p$. And, let $K = \ker \pi_H$, π_H is an action above.

Since $K \leq H$, let $|H:K| = k$. Then, $|G:K| = |G:H| \cdot |H:K| = p \cdot k$.

Since H has p left cosets, $G/K \cong \pi_H[G] \leq S_p$, so $p \cdot k = |G/K|$ divides $p!$,

so $k = 1$, $H = K$.

Automorphism of Group.

Prop. Let H be a normal subgroup of the group G .

Then, G acts by conjugation on H as automorphisms of H .

More specifically, for each $g \in G$,

$$H \rightarrow H: h \mapsto ghg^{-1}$$

is an automorphism of H . Moreover, the permutation represent,

$$\tau_H: G \rightarrow S_H: g \mapsto \sigma_g \quad \text{where } \sigma_g(h) = ghg^{-1}$$

is a homomorphism of G into $\text{Aut}(H) \leq S_H$, with $\ker \tau_H = C_G(H)$.

In summary, $G/\ker \tau_H = G/C_G(H) \cong \text{Im } \tau_H \leq \text{Aut}(H)$, and $G/Z(G) \cong \text{Inn}(G)$.

Proof. Since H is normal, the action is well-defined. And,

$$\ker \tau_H = \{g \in G \mid \sigma_g = \text{id}\} = \{g \in G \mid ghg^{-1} = h \text{ for all } h \in H\} = C_G(H).$$

Corollary. For any subgroup H of G ,

$N_G(H)/C_G(H)$ is isomorphic to a subgroup of $\text{Aut}(H)$.

Prop. $\text{Aut}(\mathbb{Z}_n) \cong \mathbb{Z}_n^\times$.

Proof. Denote $\mathbb{Z}_n = \langle x \rangle$, where $x \in \mathbb{Z}_n$. Given automorphism $\varphi \in \text{Aut}(\mathbb{Z}_n)$, the φ is uniquely determined by $\varphi(x) = x^a$, the integer $1 \leq a \leq n$, and $|x| = |\varphi(x)| = |x^a|$, so $\gcd(a, n) = 1$. Now,

$$\text{Aut}(\mathbb{Z}_n) \rightarrow \mathbb{Z}_n^\times; \varphi_a \mapsto a \pmod n$$

is the isomorphism.

$$\mathbb{Z}_8^\times = \{1, 3, 5, 7\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

Note: If p is a group of a prime order p , then

$$\text{Aut}(p) \cong \mathbb{Z}_p^\times \cong \mathbb{Z}_{p-1}.$$

Lemma. Let $P \in \text{Syl}_p(G)$. If Q is p -subgroup of G , then $Q \cap N_G(P) = Q \cap P$.

Proof. Since $P \leq N_G(P)$, $Q \cap P \leq Q \cap N_G(P)$ is trivial.

Since $Q \cap N_G(P) \leq N_G(P)$, denote $H = Q \cap N_G(P)$,

$$PH = \bigcup_{h \in H} Ph = \bigcup_{h \in H} hP = HP$$

Thus $PH \leq G$ and $|PH| = \frac{|P| \cdot |H|}{|P \cap H|}$. By the Lagrange's Thm, $H \leq Q$ means

order of H is power of p . Therefore, $P \leq PH$ and $|PH|$ is power of p , thus $P \cap H = P$, or $H \leq P$. Now, $H = Q \cap N_G(P) \leq Q$, $H \leq Q \cap P$. $\square$

Lemma. If G is finite abelian group and p is a prime number dividing $|G|$, then G has an element of order of p .

Proof. Using induction for $|G|$. If $|G|=1$, trivial.

Now, suppose that the statement hold for all groups whose order is smaller than $|G|$.

Since $|G|>1$, take $x \in G$ with $x \neq 1$. That is, $|x|>1$.

If $|G|=p$, then $|x|=p$, by the Lagrange's Theorem ($|x| \mid p$).

Now, suppose $|G| \neq p$ ($|G|>p$).

If p divides $|x|$, $|x|=p^n$ for some $n \in \mathbb{N}$, so $|x^p|=p$. So clear.

If p does not divide $|x|$, let $N = \langle x \rangle$. (Clearly, $p \nmid |N| = |x|$).

Since G is abelian, N is Normal, and $x \neq 1$, so it is non-trivial.

By the Lagrange Theorem, $|G| = |G/N| \cdot |N|$. But, p does not divide $|N|$,

p divides $|G/N|$. Now, using the hypothesis of the induction

G/N has an element of order p , denote this $\bar{y} = yN$. Since $\bar{y} \neq \bar{1}$, $y \notin N$.

But, $\bar{y}^p = \bar{1}$, so $y^p \in N$. This implies $\langle y^p \rangle \neq \langle y \rangle$. Since $|y^p|$ divides $|y|=p$, the only possibilities are $|y^p|=1$, this y is we desired.

Theorem.

Let G be a group of order $p^\alpha \cdot m$ where p^α is a power of prime p , and $p \nmid m$.

1st sylow theorem. Sylow p -subgroup of G exists.

2nd sylow theorem. If $P \in \text{Syl}_p(G)$ and Q is p -subgroup of G ,
then $Q \leq gPg^{-1}$ for some $g \in G$.

3rd sylow theorem. $n_p \equiv 1 \pmod p$ and $n_p = |G : N_G(P)|$ for any $P \in \text{Syl}_p(G)$.

Proof. To show the Existence of Sylow p -subgroup, using induction for $|G|$.

If $|G|=1$, then $p^\alpha=1$, so $\text{Syl}_p(G) = \{G\} \neq \emptyset$.

Assume that the sylow p -subgroup always exists for any group of order smaller than G .

If order of $Z(G)$ is divided by p , then by the Cauchy's Theorem for abelian group, there exists order of p element $y \in Z(G)$. Let $N = \langle y \rangle$. Denote $\overline{G} = G/N$.

By the hypothesis and the Lagrange's Theorem, $|\overline{G}| = |G|/|N| = p^{\alpha-1} \cdot m$.

By the induction, $\overline{G}$ has a sylow p -subgroup $\overline{P}$ of order $p^{\alpha-1}$.

By the correspondence isomorphism Theorem, there is $P \leq G$ such that $\overline{P} = P/N$.

By the Lagrange's Theorem again, $p^{\alpha-1} = |\overline{P}| = |P/N| = |P|/p$, so $|P| = p^\alpha$.

Therefore, there exists a sylow p -subgroup P , in case of $p \mid |Z(G)|$.

If order of $Z(G)$ is not divided by p , let $g_1, \dots, g_r$ be representatives of distinct non-central conjugacy classes of G . The class equation gives:

$$|G| = |Z(G)| + \sum_{i=1}^r |G : C_G(g_i)|$$

If all $|G : C_G(g_i)|$ are divided by p , then $|Z(G)|$ must be divided by p , contradiction.

Therefore, for some $1 \leq i \leq r$, p does not divide $|G : C_G(g_i)|$. Let $H = C_G(g_i)$.

By the Lagrange's theorem, $|H|$ divides $|G|$, so $|H| = p^\alpha \cdot k$ where $p \nmid k$, $k \mid m$.

(If $|H|$ does not have factor of p^α , then p divides $|G : C_G(g_i)|$, which is contradiction.)

And, since $g_i \notin Z(G)$, $H < G$. Therefore, by the assumption of induction,

H has a sylow p -subgroup, so it is sylow p -subgroup of G .

To show the second and third assertion, for group G of order $p^a m$, there exists a Sylow p -subgroup P of G , by the first assertion. Let

$$S = \{gPg^{-1} \mid g \in G\} = \{P_1, P_2, \dots, P_r\}$$

and let Q be any p -subgroup of G .

Let Q acts by conjugation on S . Write S as a disjoint union of orbit:

$$S = \mathcal{O}_1 \cup \dots \cup \mathcal{O}_s \quad \text{where } |\mathcal{O}_1| + \dots + |\mathcal{O}_s| = r$$

(Note that S has only one orbit under G , but it may have more than one under Q)

Reorder the elements of S so that the first s numbers elements of S are

representatives of the Q -orbit $\mathcal{O}_i$: $P_i \in \mathcal{O}_i$. $(P_1, P_2, \dots, P_s, P_{s+1}, \dots, P_r)$
 It follows that $|\mathcal{O}_i| = |Q : N_Q(P_i)|$.
 $(\underbrace{P_1, P_2, \dots, P_s}_{\mathcal{O}_1}, \underbrace{P_{s+1}, \dots, P_r}_{\mathcal{O}_2, \dots, \mathcal{O}_s})$

By definition, $N_Q(P_i) = N_G(P_i) \cap Q$ and
 by the Lemma above, $= P_i \cap Q$.

Therefore, $|\mathcal{O}_i| = |Q : P_i \cap Q|$, $1 \leq i \leq s$. Since Q was arbitrary, take $Q = P_1$. Then
 $|\mathcal{O}_1| = |Q : Q \cap Q| = 1$. Now, for $2 \leq i \leq s$, $P_1 \cap P_i < P_1$, so

$$|\mathcal{O}_i| = |P_1 : P_1 \cap P_i| > 1 \quad \text{for all } 2 \leq i \leq s.$$

Since P_1 has order of power of p , $|\mathcal{O}_i|$ must be power of p . Therefore,

$$r = \underbrace{|\mathcal{O}_1|}_{1} + \underbrace{|\mathcal{O}_2| + \dots + |\mathcal{O}_s|}_{\text{order of } p} \equiv 1 \pmod{p}. \quad \dots (*)$$

Let Q be any p -subgroup of G . Suppose that Q is not contained in P_i for all i .

Then, $Q \cap P_i < Q$ for all $1 \leq i \leq r$, therefore $|\mathcal{O}_i| = |Q : Q \cap P_i| > 1$ for each $1 \leq i \leq s$.

Since Q has order of power of p , so $|\mathcal{O}_i| > 1$ is power of p ,

therefore all $|\mathcal{O}_i|$ ($1 \leq i \leq s$) are divided by p , it contradicts to $(*)$.

Consequently, $Q \leq gPg^{-1}$ for some $g \in G$, which proves the second.

Let Q be any Sylow p -subgroup of G . By the argument above, $Q \leq gPg^{-1}$

for some $g \in G$. And since $|Q| = |gPg^{-1}|$, $Q = gPg^{-1}$.

Therefore, all Sylow p -subgroups are conjugate. Finally, for any $p \in \text{Syl}_p(G)$,

$$\# \text{Syl}_p(G) \stackrel{\text{def}}{=} n_p = |G : N_G(p)|, \quad \text{since } \text{Syl}_p(G) \text{ is the orbit.}$$

Moreover, $m = |G : p| = |G : N_G(p)| \cdot |N_G(p) : p| = n_p \cdot |N_G(p) : p|$, so $n_p \mid m$. $\square$

Corollary.

[Group of order of pg ($p < g$ are prime, $p \nmid g-1$).

If $Z(G) \neq 1$, then $G/Z(G) \cong \mathbb{Z}_p$ or $\mathbb{Z}_g$ by the Lagrange Theorem, then G must be abelian, since $G/Z(G)$ is cyclic.

Now, suppose that $Z(G) = 1$.

By the Lagrange Thm, every non-identity element of G has order p or g .

If there is no order g element, then the class equation gives

$$pg = 1 + \sum [G : C_G(g_i)] = 1 + kg,$$

because $C_G(g_i) = \langle g_i \rangle \cong \mathbb{Z}_p$. (Since $g_i \notin Z(G)$, $C_G(g_i) \neq \langle g_i \rangle$).

This is contradiction, since $g \nmid 1$. So, G has a order g element, denote this to x .

Let $H = \langle x \rangle$. Since $[G : H] = p$ is the smallest prime dividing $|G|$, so H is normal.

Now, $G/H = N_G(H)/C_G(H)$ is a subgroup of order p , and isomorphic to a subgroup of $\text{Aut}(H)$.

Since $\text{Aut}(H) \cong \mathbb{Z}_g^\times$ has order of $g-1$, this implies $p \nmid g-1$, contradiction.

[Group of order of pg , $p < g$. (p, g are prime)

Let $P \in \text{Syl}_p(G)$ and $Q \in \text{Syl}_g(G)$.

Since n_g divides P , $n_g = 1$ or P , but $n_g \equiv 1 \pmod{g}$, thus $n_g = 1$. $\therefore Q \trianglelefteq G$.

↳ Case of $P \nmid g-1$.

By the Sylow Thm, $n_p = 1$ or g , but if $n_p = g$, then $g \equiv 1 \pmod{p}$, $p \mid g-1$, contradiction.

Therefore, $n_p = 1$, $P \trianglelefteq G$. Now, G must be a cyclic, because:

Since $|P|$ and $|Q|$ are primes, both are cyclic. Let $P = \langle x \rangle$ and $Q = \langle y \rangle$.

Since $P \trianglelefteq G$, $N_G(P) = G$. Therefore,

$$G/C_G(P) = N_G(P)/C_G(P) \cong S \text{ for some } S \leq \text{Aut}(\mathbb{Z}_p)$$

But, $|\text{Aut}(\mathbb{Z}_p)| = p-1$, thus S is trivial. (because $|S|$ divides $|G| = pg$).

Now, $G = C_G(P)$, and $x \in P$, x and y are commute. This implies $|xy| = pg$. $G = \langle pg \rangle$.

2. Case of $P \mid g-1$

Let Q be a Sylow g -subgroup of S_g , the symmetric group of letters.

Since g is prime, $|N_{S_g}(Q)| = g \cdot (g-1)$, because

1). Since $|S_g| = g!$, $|Q| = g$.

2). S_g has $(g-1)!$ g -cycles.

3). Since order of gQg^{-1} is g , these are all cyclic.

4). $gQg^{-1} \cap Q = \{1\}$ for all $g \in G$, so

$$n_g = \# \{gQg^{-1} \mid g \in G\} = (g-1)! / (g-1) = (g-2)!$$

$$\text{Therefore } n_g = G_Q = [S_g : N_G(Q)] = (g-2)!, \text{ or } N_{S_g}(Q) = g(g-1). \quad \overset{p}{\uparrow}$$

Now, since $p \mid g-1$, p divides order of $N_{S_g}(Q)$, so it has a subgroup of order p .

And, since $p \leq N_{S_g}(Q)$, pQ is subgroup of order pg .

And, $C_{S_g}(Q) = Q$, pQ is non-abelian group.

$$|G : N_{S_g}(Q)| = G_Q$$

Group of order 48 is not simple.

$48 = 2^4 \cdot 3$. Therefore,

$$n_3 = 1 \text{ or } 4 \text{ or } 16 \quad \text{and} \quad n_2 = 1 \text{ or } 3.$$

If $n_2 = 1$, then $\text{PG Syl}_2(G)$ is the Normal, so there is nothing to prove.

Suppose $n_2 \neq 1$, or $n_2 = 3$. That is, there are three Sylow 2-subgroup,

$$P_1, P_2, P_3 \in G. \quad (\#P_i = 16).$$

For each $1 \leq i < j \leq 3$,

$$|P_i P_j| = \frac{|P_i| \cdot |P_j|}{|P_i \cap P_j|} = \frac{2^8}{|P_i \cap P_j|} \leq 48 = |G| \quad \text{if } i, j, P_i = P_j$$

So, $\frac{2^4}{3} \leq |P_i \cap P_j|$, $|P_i \cap P_j|$ cannot be 1, 2, 4, 16 therefore

$$|P_i \cap P_j| = 8.$$

Take $Q = P_1 \cap P_2$, as subgroup of order 8. Since $Q \subseteq P_1, P_2$ and index 2, $Q \trianglelefteq P_1$ and $Q \trianglelefteq P_2$. That is, for any $g \in P_1 \cup P_2$,

$$gQg^{-1} = Q.$$

therefore, $P_1 \cup P_2 \subseteq N_G(Q)$. Since $|P_1 \cap P_2| = 8$,

$$|P_1 \cup P_2| = |P_1| + |P_2| - |P_1 \cap P_2| = 24$$

Now, if $|N_G(Q)| = 24$, then it has index 2 to G ,

$$N_G(Q) \trianglelefteq G$$

And, if $|N_G(Q)| = 48$, then $N_G(Q) = G$, so $Q \trianglelefteq G$.

[Group of order 30 is not simple.

$30 = 2 \cdot 3 \cdot 5$, let $P \in \text{Syl}_5(G)$ and $Q \in \text{Syl}_3(G)$.

If either P or Q is normal, then $PQ \leq G$, and it is group of order 15, because $|PQ| = \frac{|P| \cdot |Q|}{|P \cap Q|} = 15$. (All elements of P has order 5 except identity, and " Q " 3 " 1).

So, PQ is index of 2 to G , so PQ is normal subgroup.

Now, assume neither P nor Q is normal. By the Sylow Theorem,

$n_5 = 6$ and $n_3 = 10$. By the Lagrange's Theorem, for any $P_1, P_2 \in \text{Syl}_5(G)$,

$P_1 \cap P_2 = \{1\}$. So, there are $6 \cdot 4 = 24$ distinct order 5 elements.

Similarly, there are 20 distinct order 3 elements, $24 + 20 = 44 \leq 30$,

Contradiction. Therefore, either P or Q must be normal, so PQ is index 15 subgroup.

In sum. Group of order 30 must have a subgroup of order of 15.

[Group of order 105 is Not simple.

$105 = 3 \cdot 5 \cdot 7$, so

$$n_3 = 1 \text{ or } 7$$

$$n_5 = 1 \text{ or } 21$$

$$n_7 = 1 \text{ or } 15$$

If it is simple, then $n_3 = 7$, $n_5 = 21$, $n_7 = 15$. Therefore there are

$$\underbrace{2 \cdot 7}_{\text{order 3}} + \underbrace{4 \cdot 21}_{\text{order 5}} + \underbrace{6 \cdot 15}_{\text{order 7}} = 14 + 84 + 90 \geq 130$$

So, Contradiction by Counting

[Group of order 12 is Not Simple.

Every group of order 12 has a normal sylow 3-subgroup or $G \cong A_4$.

$12 = 2^2 \cdot 3$. If $n_3 = 1$, then G is not simple, it has a normal sylow 3-subgroup.

Suppose that $n_3 \neq 1$. Then, there is only possibility is $n_3 = 4$.

Let $P \in \text{Syl}_3(G)$. By the sylow Theorem, $|G/N_G(P)| = n_3 = 4$,

so $P = N_G(P)$, by $P \leq N_G(P)$ and $|N_G(P)| = 12/4 = 3$, given by the Lagrange Thm.

Now, consider an action of G on $\text{Syl}_3(G)$ by conjugation:

$$G \times \text{Syl}_3(G) \rightarrow \text{Syl}_3(G) : (g, P) \mapsto gPg^{-1}$$

Since $\text{Syl}_3(G) = \{P_1, P_2, P_3, P_4\}$, we can write the permutation representation of this action:

$$\mathcal{Q}: G \rightarrow S_4 : g \mapsto \sigma_g \quad \text{where} \quad \sigma_g: S_4 \rightarrow S_4 : P_i \mapsto gP_i g^{-1}$$

Now, the elements of $\ker \mathcal{Q}$ normalizes all sylow 3-subgroups. That is,

$$\ker \mathcal{Q} \leq N_G(P) = P$$

Since P is not normal in G , $\ker \mathcal{Q} = 1$. That is, $\mathcal{Q}$ is injective,

$$G \cong \mathcal{Q}[G] \leq S_4.$$

Since G contains 8 elements of order 3, by counting,

and S_4 has exactly 8 element of order 3, and all these are even,

Thus $G \cong A_4$.

[Groups of $p^2 \cdot q$, p and q distinct primes.

Let G be a group of order $p^2 \cdot q$. Let $P \in \text{Syl}_p(G)$ and $Q \in \text{Syl}_q(G)$.

[In case of $p > q$, then $n_p = 1$ or $1+kp$ ($k \geq 1$), but $1+kp \nmid q$, so $n_p = 1$.

Thus $P \trianglelefteq G$.

□ If $|G|=60$ and G has more than one Sylow 5-subgroup, then G is simple.

Proof. $60 = 5!$. Since $n_5 > 1$, then only possibility is $n_5 = 6$.

Assume there is non-trivial proper normal $H \triangleleft G$ exists.

If $5 \mid |H|$, then it has at least one Sylow-5 subgroup, and since $H \triangleleft G$, all Sylow 5-subgroups are contained in H , so,

$$|H| \geq 1 + 4 \cdot 6 = 25,$$

therefore $|H| = 30$. but, since order of 30 has a unique normal Sylow 5-subgroup, so contradiction. Thus, $5 \nmid |H|$.

If $|H| = 6$ or 12 ,